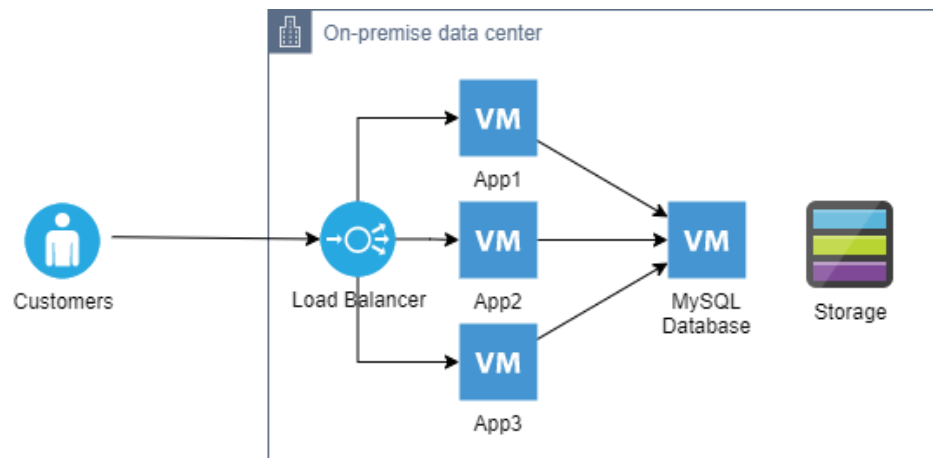


Amazon AWS I – Cloud Practitioner

Final Project

My Healthy Eating Inc., a Toronto-based company, sells healthy food online to customers across North America. The company builds and maintains its own e-Commerce website. Over the last few years, the company has experienced rapid growth due to the increasing popularity of their products. Meanwhile, their IT operational cost has also gone up significantly.

Below is a diagram of the company's on-premise data center environment, which is a typical three-tier application, including a load balancer, three application servers running the e-Commerce application, and a MySQL database server. In addition, the company has a total of approximately 2 TB data stored in the data center.



As an AWS certified cloud practitioner, you decide to write a report to the IT manager suggesting the company to migrate its on-premise data center to AWS. In the report, you need to discuss the cloud benefits, identify the AWS services to use and high-level cost estimation, and outline a brief migration plan.

Write your report below:

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Recommendation to migrate the e-commerce system to AWS

Date: 17 July 2026

1. Executive Summary

My Healthy Eating Inc. currently runs its e-commerce platform from a Toronto data centre. The existing setup includes a load balancer, three application servers, a MySQL database, and about 2 TB of stored data. As the business has grown, the cost and effort required to maintain this environment have also increased.

Moving the system to AWS would give the company a more flexible and reliable platform. It would reduce the need to maintain physical hardware, make it easier to handle changes in customer demand, and improve the availability of the online store.

2. Benefits of Migrating to AWS

One of the biggest advantages of AWS is **scalability**. Customer traffic may rise during holidays, promotions, or new product launches. Instead of buying extra hardware that may sit unused most of the time, Amazon EC2 Auto Scaling can add servers when traffic increases and remove them when demand drops. This helps maintain performance while keeping costs under better control. ([AWS Documentation](#))

AWS would also improve **availability and reliability**. The application could run across more than one Availability Zone, which reduces the risk of a complete outage if one location has a problem. An Application Load Balancer would direct traffic to healthy servers, while Amazon RDS Multi-AZ could maintain a standby MySQL database for failover. ([AWS Documentation](#))

Another benefit is that AWS can reduce routine administration. Amazon RDS handles tasks such as provisioning, backups, patching, monitoring, and recovery. This would allow the IT team to spend less time maintaining database infrastructure and more time improving the company's e-commerce system. ([Amazon Web Services, Inc.](#))

AWS also follows a pay-as-you-go model. The company would pay for the resources it uses instead of purchasing hardware in advance. Services such as AWS Budgets, Cost Explorer, and CloudWatch could help the company monitor usage and control spending.

Other benefits include faster deployment, easier backup and recovery, stronger security controls, centralized monitoring, and simpler expansion across North America.

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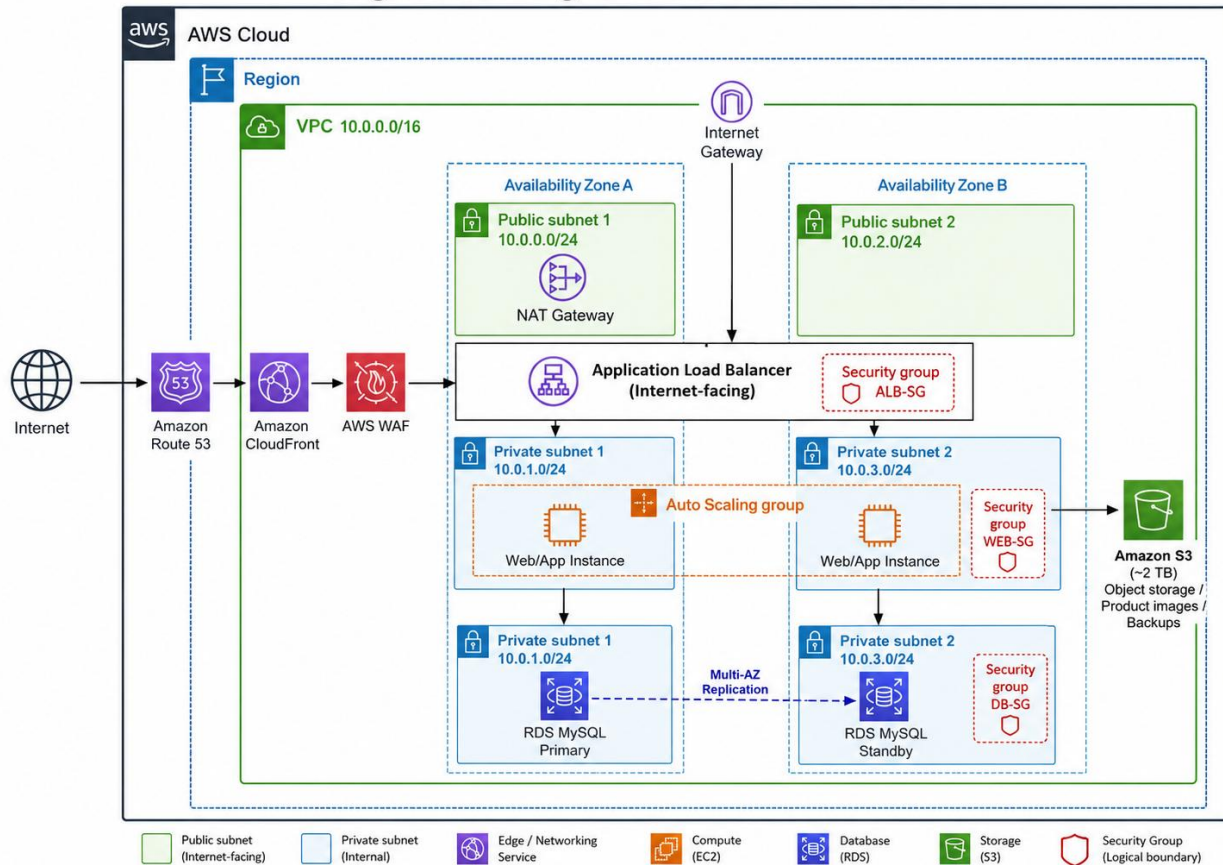
3. Recommended AWS Services

The existing three-tier design can be moved to AWS without requiring a complete redesign.

Requirement	Recommended AWS service
Secure cloud network	Amazon VPC
Customer traffic distribution	Application Load Balancer
Application hosting	Amazon EC2
Automatic capacity adjustment	EC2 Auto Scaling
Managed MySQL database	Amazon RDS for MySQL
Database failover	RDS multi-AZ
Approximately 2 TB of files	Amazon S3
EC2 disk storage	Amazon EBS
Faster content delivery	Amazon CloudFront
Domain name routing	Amazon Route 53
Access and application security	IAM, security groups and AWS WAF
Monitoring and auditing	Amazon CloudWatch and AWS CloudTrail
Server and database migration	AWS Transform MGN and AWS DMS

The Application Load Balancer would be internet-facing, while the EC2 application servers and RDS database would be protected in private subnets. The EC2 servers would run across two Availability Zones and be managed by an Auto Scaling group. Amazon S3 could store product images, files, and backups.

High-Level Target Architecture on AWS



4. High-Level Cost Estimate

The actual monthly cost would depend on traffic, server size, database usage, backup retention, storage, and outbound data transfer. The AWS Pricing Calculator should be used before migration to create a more accurate estimate. ([Amazon Web Services, Inc.](https://aws.amazon.com/pricing/))

For a small-to-medium production environment using multiple EC2 instances, an Application Load Balancer, Auto Scaling, Multi-AZ RDS, monitoring, backups, and about 2 TB of storage, a reasonable estimate would be:

Approximately CAD \$1,000 to \$3,000 per month

A lightly used environment may stay near the lower end of this range. Higher traffic, larger servers, more backups, or heavy data transfer could increase the cost.

There may also be a one-time migration cost of several thousand dollars for planning, testing, staff time, application changes, and possible professional support.

5. High-Level Migration Plan

The migration should be completed in stages so that normal business operations are not interrupted.

Assessment and planning:

The IT team should first review the current servers, database, storage, application dependencies, security requirements, software licenses, traffic levels, and acceptable downtime.

AWS Environment Setup

The company should then create the AWS network, subnets, IAM permissions, security groups, monitoring, backup policies, and cost controls. A test environment should be prepared before moving the production system.

Server and Data Migration

The application servers could be moved to Amazon EC2 using AWS Transform MGN, which supports continuous replication and shorter cutover windows. ([AWS Documentation](#))

The MySQL database could be migrated to Amazon RDS using AWS Database Migration Service. The 2 TB of file data could be moved to Amazon S3 or EBS, depending on how the application uses it.

Testing

Before the final move, the company should test the website, customer accounts, orders, payments, inventory, performance, security, backups, and database failover.

Cutover

The final data changes should be synchronized during a low-traffic period. Customer traffic could then be redirected to AWS through DNS. A rollback plan should remain available in case a serious problem occurs.

Optimization

Once the AWS environment is stable, the IT team should review performance and cost, right-size EC2 and RDS resources, remove temporary migration resources, and gradually retire the old data-centre systems.

6. Important Migration Considerations

The company should pay particular attention to:

- Customer and payment-data protection

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- Encryption during transfer and storage
- PCI DSS and privacy requirements
- Backup and disaster-recovery planning
- Application and software compatibility
- DNS changes and acceptable downtime
- Staff training
- Data-transfer and long-term cloud costs
- A tested rollback and communication plan

7. Conclusion

Migrating to AWS would give My Healthy Eating Inc. a more scalable, reliable, and manageable platform for its growing e-commerce business. Services such as Amazon EC2, Auto Scaling, Application Load Balancer, Amazon RDS, and Amazon S3 could replace the current infrastructure while reducing physical hardware management.

A phased migration with proper testing, continuous data replication, a controlled cutover, and a rollback plan would reduce risk and help avoid disruption to customers. The expected monthly cost would likely be in the low thousands of Canadian dollars, depending on the company's actual usage.